

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 6: Thresholding

INPUT DATA

Given Pixel Values: 45, 60, 120, 130, 200, 210, 75, 90

Applied Threshold: $T = 100$

TASKS 1–3: BINARY CONVERSION & OUTPUT VECTOR

Using the standard thresholding rule:

$$g(x, y) = 1 \text{ if } f(x, y) \geq T, \text{ else } 0$$

Original Pixel Value	Comparison (≥ 100)	Binary Output Value
45	$45 < 100$	0
60	$60 < 100$	0
120	$120 \geq 100$	1
130	$130 \geq 100$	1
200	$200 \geq 100$	1
210	$210 \geq 100$	1
75	$75 < 100$	0
90	$90 < 100$	0

Output Binary Vector:

[0, 0, 1, 1, 1, 1, 0, 0]

TASK 4: WHAT HAPPENS IF THE THRESHOLD IS INCREASED?

If the threshold value T is increased (e.g., set to $T = 150$):

- **Fewer pixels become 1 (Foreground):** Pixels in the intermediate range (such as 120 and 130) drop below the new higher threshold and shift from 1 to 0.
- **More pixels become 0 (Background):** The number of background pixels increases while foreground features shrink.
- **Effect on Image Output:** Higher threshold values lead to smaller, stricter foreground segments, which risks losing weaker or dimmer foreground components.